HistoLens, a graphical user interface (GUI) facilitates wholescale analysis of histology image datasets. By allowing the user to dynamically interact with their data, *HistoLens* allows for the investigation of highly specific hypotheses.

HistoLens identifies distinct patterns of podocyte injury in HIV-transgenic mice

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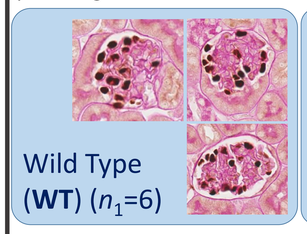
Motivation

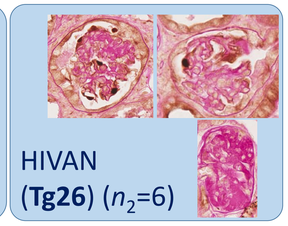
The field of computational pathology is currently experiencing rapid amounts of growth. The application of complex machine learning (ML) models to histological data enables researchers to automate simple diagnostic and region of interest (ROI) detection tasks on a large scale. However, the primary disadvantage of these techniques is that they are not always accessible to pathologists. Additionally, relying on ML models to perform diagnoses does not provide pathologists with usable information from which additional treatments can be designed. Through the use of hand-engineered features and the direct linkage of each data point to the original image, *HistoLens* enables pathologists to interpret quantitative characteristics of annotated structures.

Dataset

HIV associated nephropathy (HIVAN) is a major cause of End-Stage Renal Disease (ESRD), particularly for individuals of African descent. Histological features of HIVAN include mesangial cell proliferation, collapsing focal segmental glomerulosclerosis, and podocyte injury. Podocytes, a terminally differentiated epithelial cell typically found on outer surfaces of glomerular capillaries, fill a critical role in the filtration of proteins from the bloodstream. Assessing the extent of podocyte injury across several whole slide images (WSIs) is an exceptionally difficult task, even for experienced pathologists.



Wild Type (**WT**) (n_1=6)

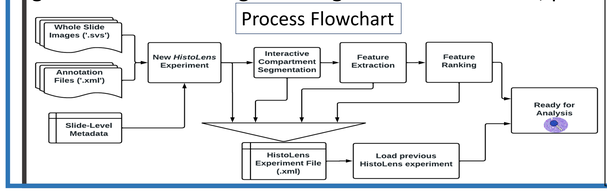


HIVAN (**Tg26**) (n_2=6)

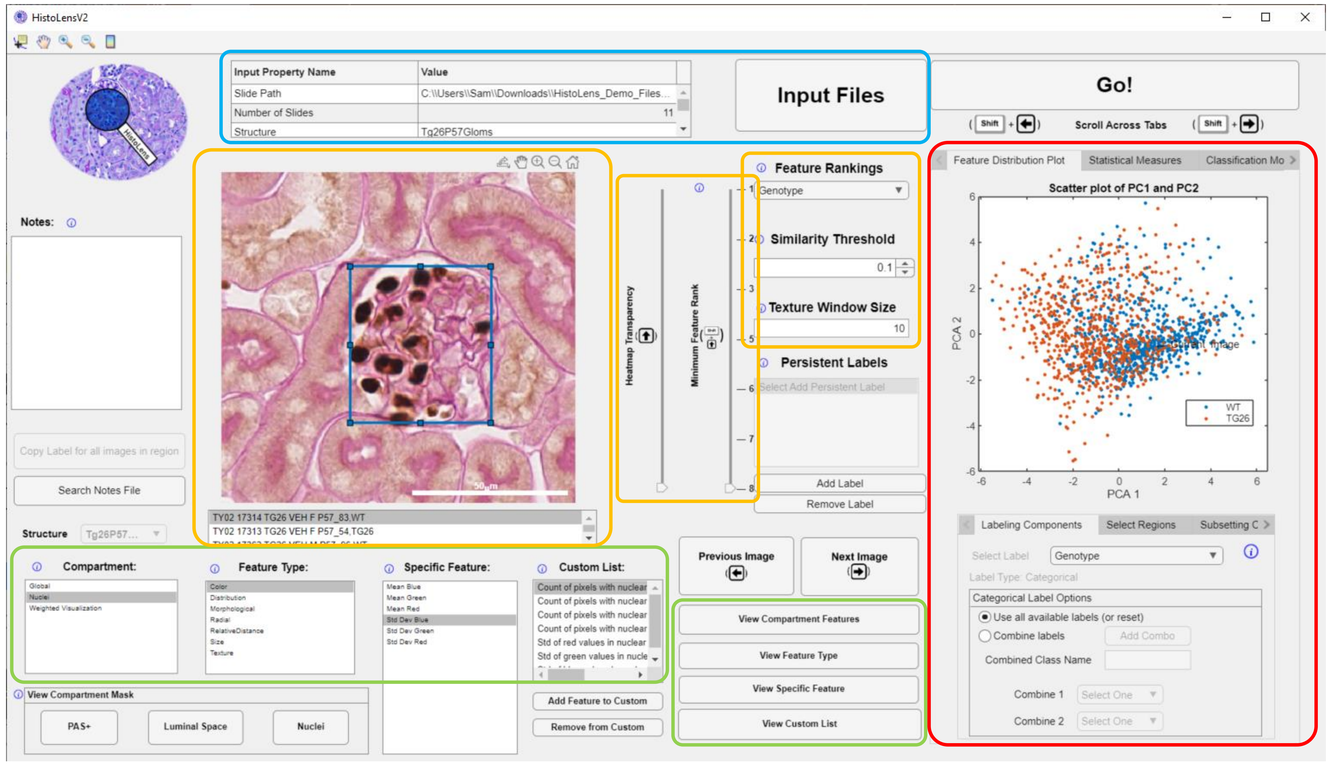
Hand-engineered features

Quantitative features integrated into *HistoLens* are based on the glomerular features introduced by *Ginley et al*. They include measures of morphology, color, texture, and relative distribution within the glomerular capsule. Podocytes were isolated from other nuclei in the glomerular tuft through staining of the CDK-inhibitor, p57.

Process Flowchart



HistoLens main window



Input Files

Input path to WSIs and go to feature extraction or input experiment file to reload previous analysis

Image Viewing

Select specific image to view and adjust transparency of feature visualization

Select Features

Choose between specific quantitative image features or groups of quantitative features to view distribution in Feature Distribution Plot

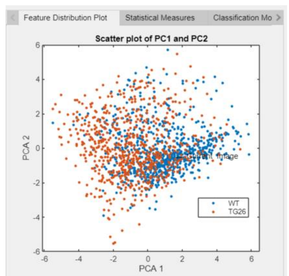
Analysis Toolbox

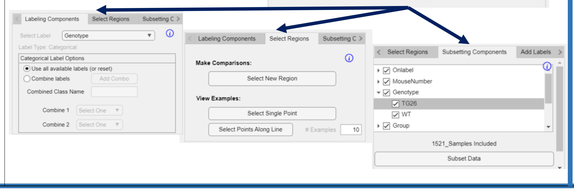
Carry out a variety of different analytical tasks including changing labels used in Feature Distribution Plot, viewing statistical relationships, training classifiers, and recording annotations

Analytics Toolbox highlights

Feature Distribution Plot:

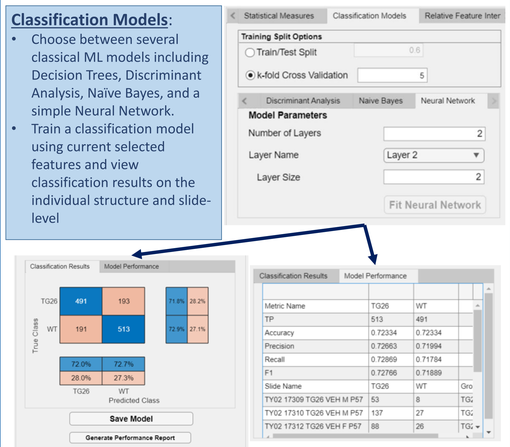
- View distribution of selected feature or group of features
- Additional functionality includes changing how points are labeled according to provided metadata, selecting specific points or regions within the distribution plot for comparisons, subsetting data according to provided metadata, adding labels from an external file, and removing outliers





Classification Models:

- Choose between several classical ML models including Decision Trees, Discriminant Analysis, Naïve Bayes, and a simple Neural Network.
- Train a classification model using current selected features and view classification results on the individual structure and slide-level

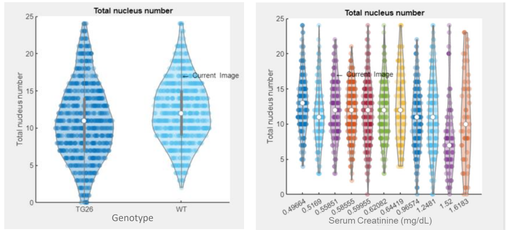


Add Annotations:

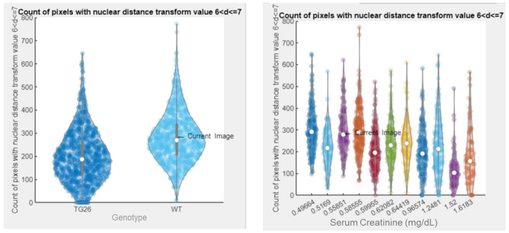
- Dynamically add classes to annotate in each image.
- Annotations saved as binary masks to be used in future analyses.

Results

Using *HistoLens*, we were able to identify several quantitative features that distinguished the HIVAN (Tg26) glomeruli from Wild-type (WT). In general, Tg26 mice had a smaller podocyte area per unit glomerular area, a higher degree of heterogeneity for RGB color values, and a lower count of pixels within podocyte distance transform ranges. These observations across our dataset confirm the presence of inflammatory stress specifically in the podocytes of Tg26 mice resulting in increased apoptosis and transcriptional activity.



Distribution of podocyte number separated first by mouse genotype (WT and Tg26) and then by Serum Creatinine (mg/dL).



Distribution of specific nuclear distance transform range pixel count separated first by mouse genotype (WT and Tg26) and then by Serum Creatinine (mg/dL).

Opportunities for contribution

We are welcoming any suggestions or offers to contribute and welcome those interested to visit the project Github repo and put in a request for specific features. In future versions of *HistoLens*, we are particularly looking to:

- Integrated WSI segmentation
- Additional structure-specific features and visualizations
- Develop disease-specific workflows to standardize analysis and increase translatability of research
- Implement additional dimensional reduction methods such as UMAP or tSNE

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